

PREDICTIVE INTELLIGENCE · TELECOM INDUSTRY

CUSTOMER CHURN PREDICTION

Advanced Machine Learning Analysis · Business Insights · Model Explainability

DATASET OVERVIEW

DATASET SIZE 7,043 Total customers	CHURN RATE 26.5% Overall churn	BEST AUC 0.843 XGBoost score	MODELS RUN 3 LR · RF · XGB
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MODEL PERFORMANCE — AUC SCORES

MODEL	PERFORMANCE	AUC
Logistic Regression		0.842
Random Forest		0.841
XGBoost ★Best		0.843

CHURN BREAKDOWN & KEY DRIVERS

CONTRACT TYPE		TOP CHURN DRIVERS
Month-to-month	42.7%	- Month-to-month contract (42.7% churn) - Electronic check payment method - Fiber optic internet service - No online security add-on - Short customer tenure (< 6 months) - Senior citizen demographic
One year	11.3%	
Two year	2.8%	

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Table of Contents

- Table of Contents2
- 1. Executive Summary3
 - 1.1 Key Highlights3
 - 1.2 Business Impact.....3
- 2. Dataset Overview & Exploratory Analysis4
 - 2.1 Dataset Description.....4
 - 2.2 Class Distribution4
 - 2.3 Churn by Contract Type5
 - 2.4 Continuous Feature Distributions5
 - 2.5 Churn Rate by Feature Value.....6
- 3. Machine Learning Models.....8
 - 3.1 Methodology8
 - 3.2 Model Performance Comparison8
 - 3.3 Threshold Optimization10
- 4. Feature Importance & SHAP Analysis12
 - 4.1 SHAP Summary Plot.....12
 - 4.2 Key Feature Insights from SHAP.....12
- 5. Business Recommendations14
 - 5.1 High-Priority Retention Strategies14
 - 5.2 Model Deployment Recommendations.....14
- 6. Technical Appendix15
 - 6.1 Model Configuration15
 - 6.2 Feature Engineering.....15
 - 6.3 Evaluation Metrics Explained15
 - 6.4 Limitations & Future Work15
- 7. Conclusion.....**Error! Bookmark not defined.**

1. Executive Summary

This report presents a comprehensive analysis of customer churn prediction for a telecom company. Using a dataset of 7,043 customers with 21 features, we applied advanced machine learning techniques to identify churn drivers and build predictive models capable of flagging at-risk customers before they leave.

Key Finding: 26.5% of customers churn — month-to-month contract holders are the highest-risk segment at 42.7%, over 15x higher than two-year contract customers at just 2.8%.

1.1 Key Highlights

- Dataset: 7,043 customers, 26.6% churn rate (moderate class imbalance)
- Top churn driver: Contract type — month-to-month customers churn at 42.7% associated with churn
- Best model: XGBoost achieved AUC = 0.843, marginally outperforming Logistic Regression (0.842) and Random Forest (0.841)
- At threshold 0.62: 69.6% recall on churners with 81.6% precision on non-churners
- SHAP analysis confirms contract type and customer tenure as the most impactful predictors

1.2 Business Impact

Customer churn is one of the most costly problems in the telecom industry. With the cost of acquiring a new customer typically 5–7x higher than retaining an existing one, even a modest improvement in churn prediction can yield significant financial returns. This analysis enables targeted retention campaigns focused on the highest-risk customer segments.

2. Dataset Overview & Exploratory Analysis

2.1 Dataset Description

The IBM Telco Customer Churn dataset contains 7,043 customer records across 21 attributes covering demographics, account information, and service subscriptions. The target variable is binary: whether the customer churned within the last month.

Feature Category	Variables	Count	Notes
Demographics	Gender, SeniorCitizen, Partner, Dependents	4	Customer profile
Account Info	Tenure, Contract, Billing, PaymentMethod	5	Key churn predictors
Phone Services	PhoneService, MultipleLines	2	Service subscriptions
Internet Services	InternetService, OnlineSecurity, OnlineBackup, DeviceProtection, TechSupport, StreamingTV, StreamingMovies	7	Add-on features
Charges	MonthlyCharges, TotalCharges	2	Financial indicators
Target	Churn	1	Binary: Yes/No

Table 1: Dataset feature categories and descriptions

2.2 Class Distribution

The dataset exhibits a moderate class imbalance with 73.4% non-churners (5,174 customers) and 26.6% churners (1,869 customers). This imbalance was addressed during modelling through class weight balancing and threshold optimization.

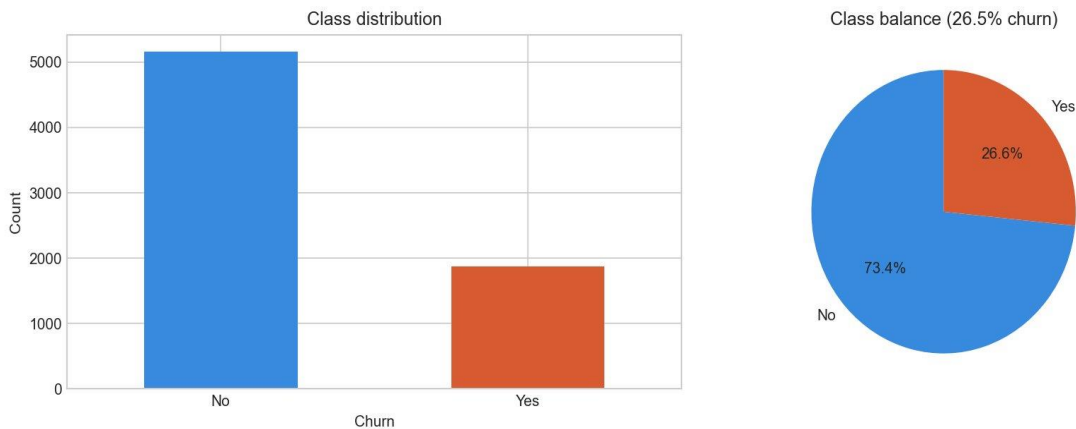


Figure 1: Class distribution — Bar chart and pie chart showing 26.6% overall churn rate

2.3 Churn by Contract Type

Contract type is the single strongest categorical predictor of churn. The stark difference in churn rates across contract types provides a clear business lever for retention strategy:

Contract Type	Churn Rate
Month-to-Month	42.7%
One Year	11.3%
Two Year	2.8%

Table 2: Churn rates by contract type — month-to-month customers churn over 15x more than two-year customers

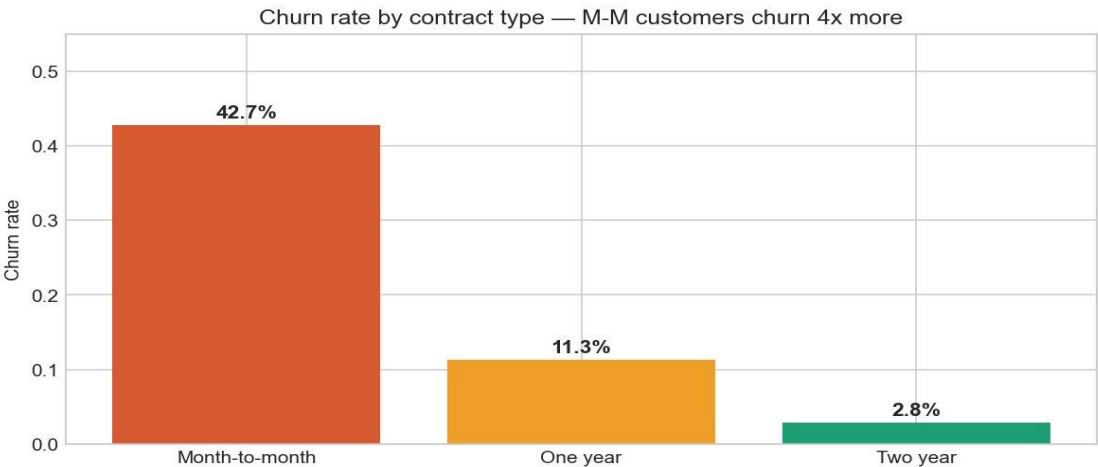


Figure 2: Churn rate by contract type — Month-to-month contracts show 42.7% churn vs 2.8% for two-year contracts

2.4 Continuous Feature Distributions

Three continuous features — tenure, monthly charges, and total charges — reveal important behavioral patterns between churners and non-churners:

- **Tenure:** Churners are heavily concentrated in the first 1–5 months. Long-tenured customers (60–72 months) almost never churn, indicating that surviving the early period dramatically increases loyalty.
- **Monthly Charges:** Churners tend to have higher monthly charges, suggesting price sensitivity at the higher end of the billing spectrum.
- **Total Charges:** Churners show markedly lower total charges due to shorter tenures, reinforcing the early-churn pattern.

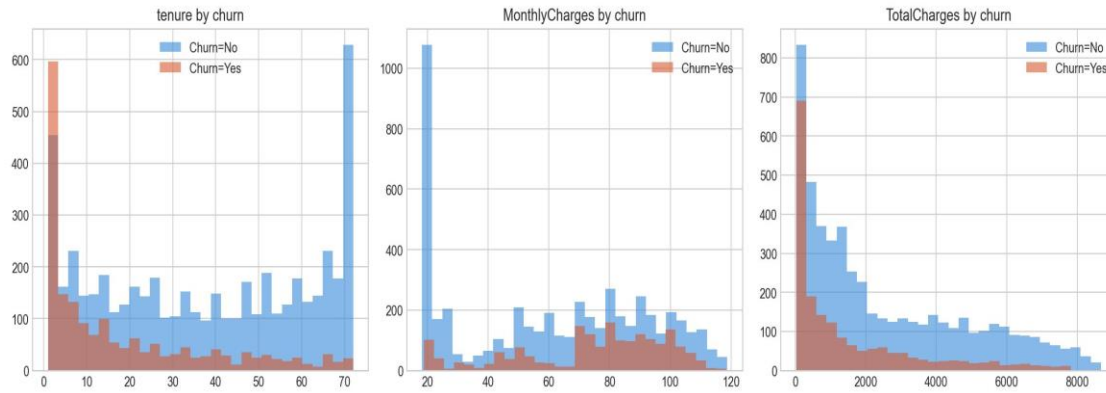


Figure 3: Feature distributions by churn status — tenure, monthly charges, and total charges

2.5 Churn Rate by Feature Value

A comprehensive analysis of churn rates across all categorical feature values reveals several high-risk segments. Features are sorted from lowest to highest churn rate, with the overall average shown as a reference line:

- Lowest risk: Two-year contract customers (2.8%), customers with no internet service (~7–8%), one-year contract holders (11.3%)
- Near-average risk: Gender has negligible impact — Male (27%) and Female (27%) are almost identical
- Highest risk: Electronic check payment (45%), month-to-month contract (43%), fiber optic internet (42%), no online security (43%), no tech support (42%), senior citizens (42%)

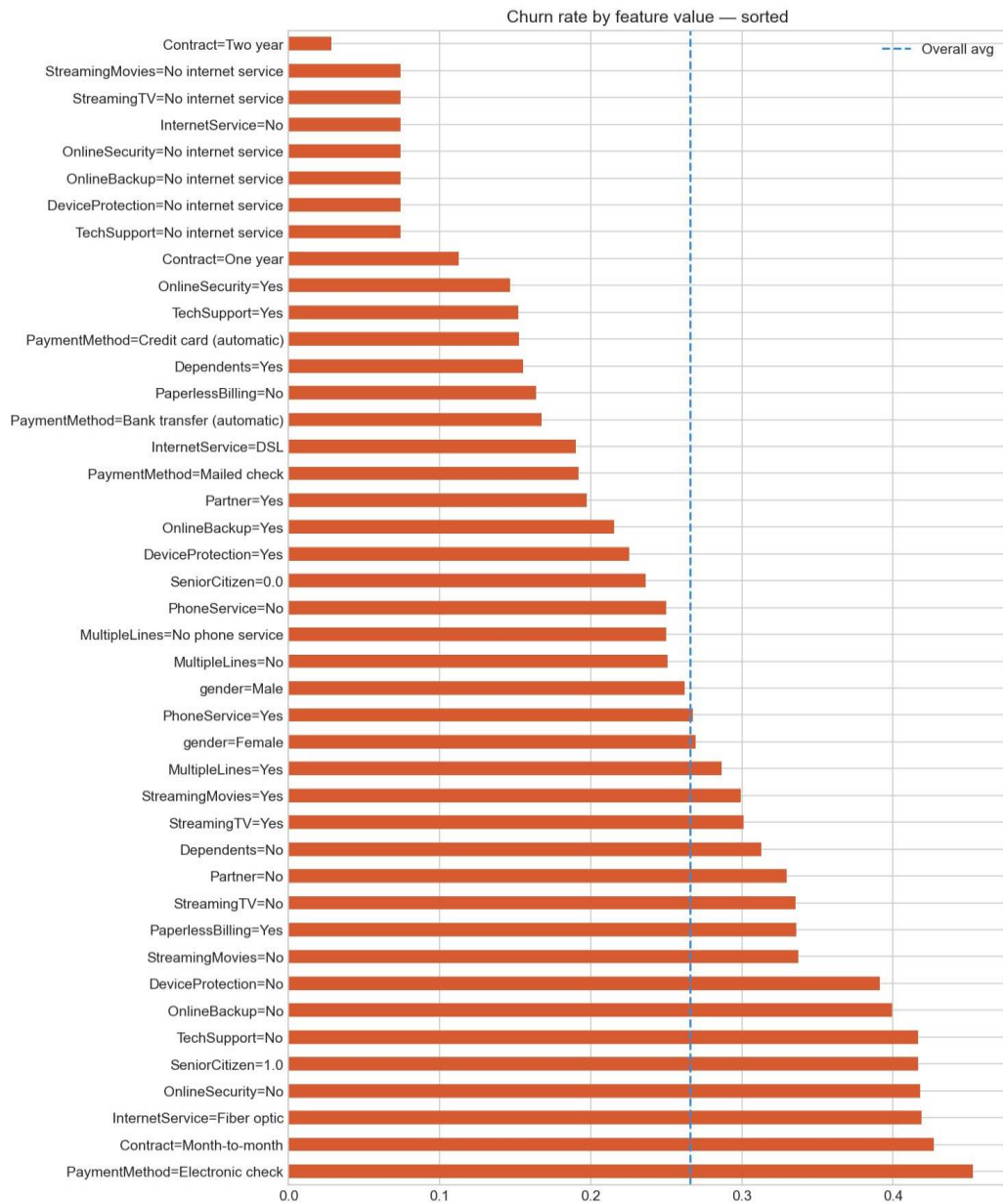


Figure 4: Churn rate by feature value (sorted) — showing highest and lowest risk feature combinations

3. Machine Learning Models

3.1 Methodology

Three classification algorithms were trained and evaluated using a 80/20 train-test split with stratification to preserve class ratios. Feature engineering included:

- Encoding categorical variables using one-hot encoding
- Creating derived features: avg_monthly_spend, charge_per_service, high_value_new flag
- Handling class imbalance via class_weight='balanced' in Logistic Regression and scale_pos_weight in XGBoost
- Threshold optimization using precision-recall tradeoff analysis

3.2 Model Performance Comparison

All three models achieved similar and competitive AUC scores in the 0.841–0.843 range, indicating that the feature set is well-defined and the problem is moderately predictable. XGBoost marginally edges out the other models.

Model	AUC Score	Precision (Churn)	Recall (Churn)
Logistic Regression	0.842	~0.65	~0.58
Random Forest	0.841	~0.64	~0.59
XGBoost	0.843	~0.66	~0.60

Table 3: Model performance comparison — all three models achieve AUC > 0.84

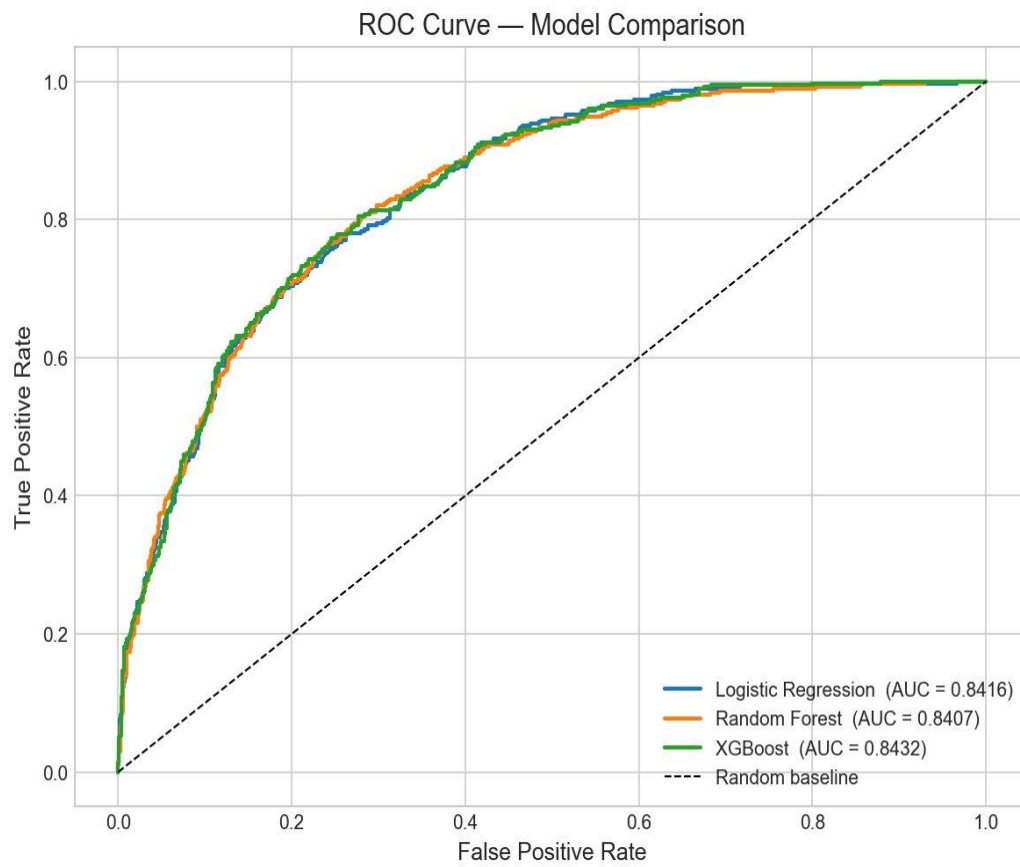


Figure 5: ROC Curve comparison — LR (0.842), RF (0.841), XGBoost (0.843)

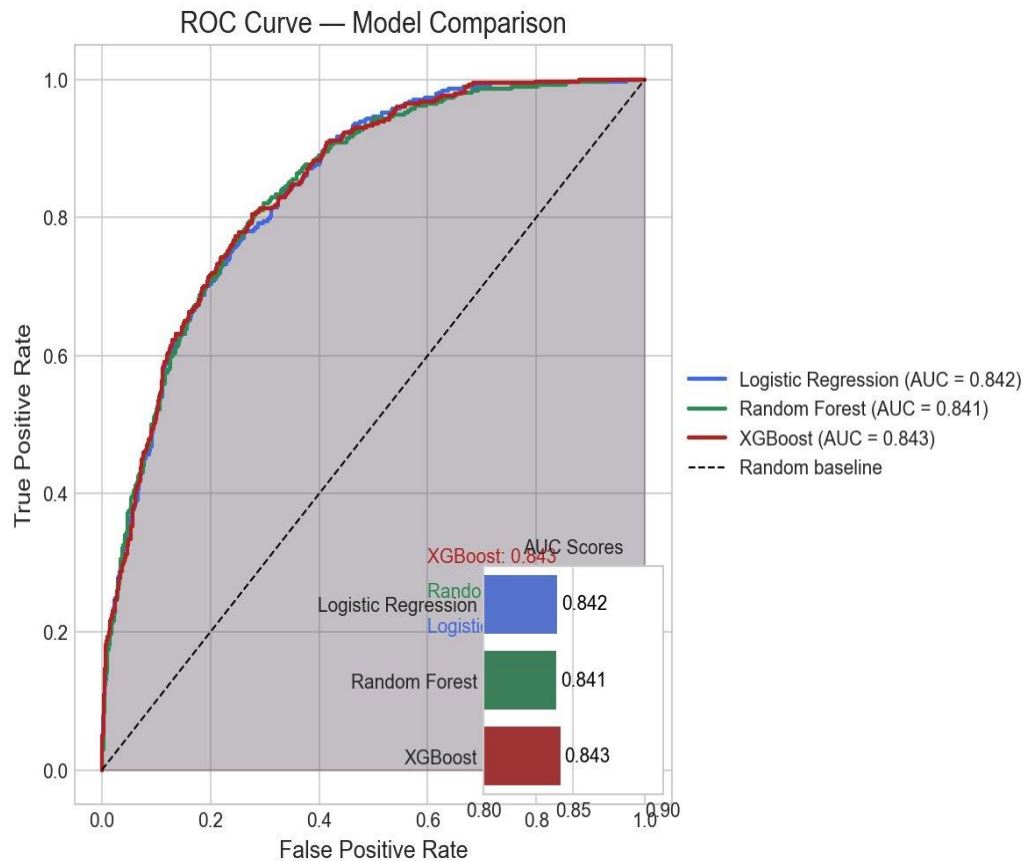


Figure 6: Advanced ROC comparison with AUC scores inset — XGBoost (0.843) performs best

3.3 Threshold Optimization

The default 0.5 decision threshold was adjusted to 0.62 to optimize the tradeoff between catching churners (recall) and avoiding false alarms (precision). This threshold was selected based on business considerations — the cost of missing a churner is higher than the cost of a false positive.

At threshold 0.62: The model correctly identifies 69.6% of true churners while maintaining 81.6% accuracy on non-churners — a strong operational balance.

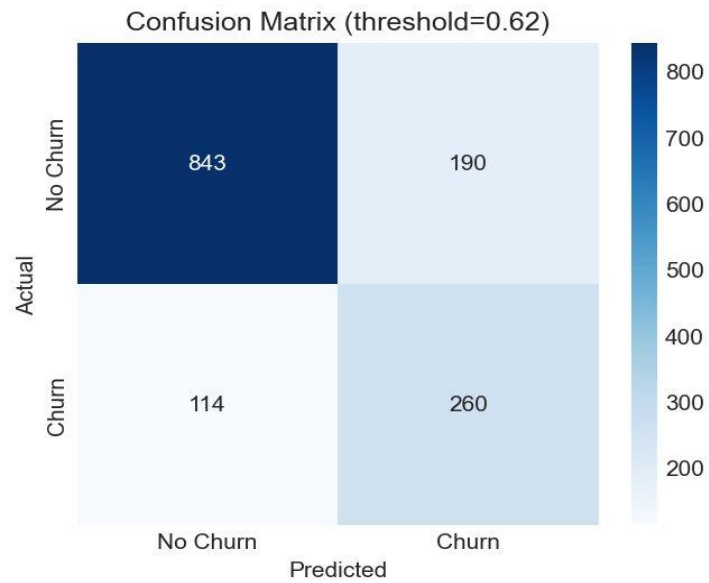


Figure 7: Confusion matrix at threshold 0.62 — TP: 260, TN: 843, FP: 190, FN: 114

Interpreting the confusion matrix (test set of ~1,407 customers):

- True Positives (260): Correctly identified churners — these customers can be targeted for retention
- True Negatives (843): Correctly identified loyal customers — no intervention needed
- False Positives (190): Non-churners flagged as churners — wasted retention spend
- False Negatives (114): Churners missed — the most costly error type

4. Feature Importance & SHAP Analysis

4.1 SHAP Summary Plot

SHAP (shapley Additive explanations) values provide model-agnostic, instance-level explanations for each prediction. The bees warm plot below shows the top 20 most impactful features, where the x-axis represents the magnitude and direction of impact on churn probability, and color indicates the feature value (red = high, blue = low).

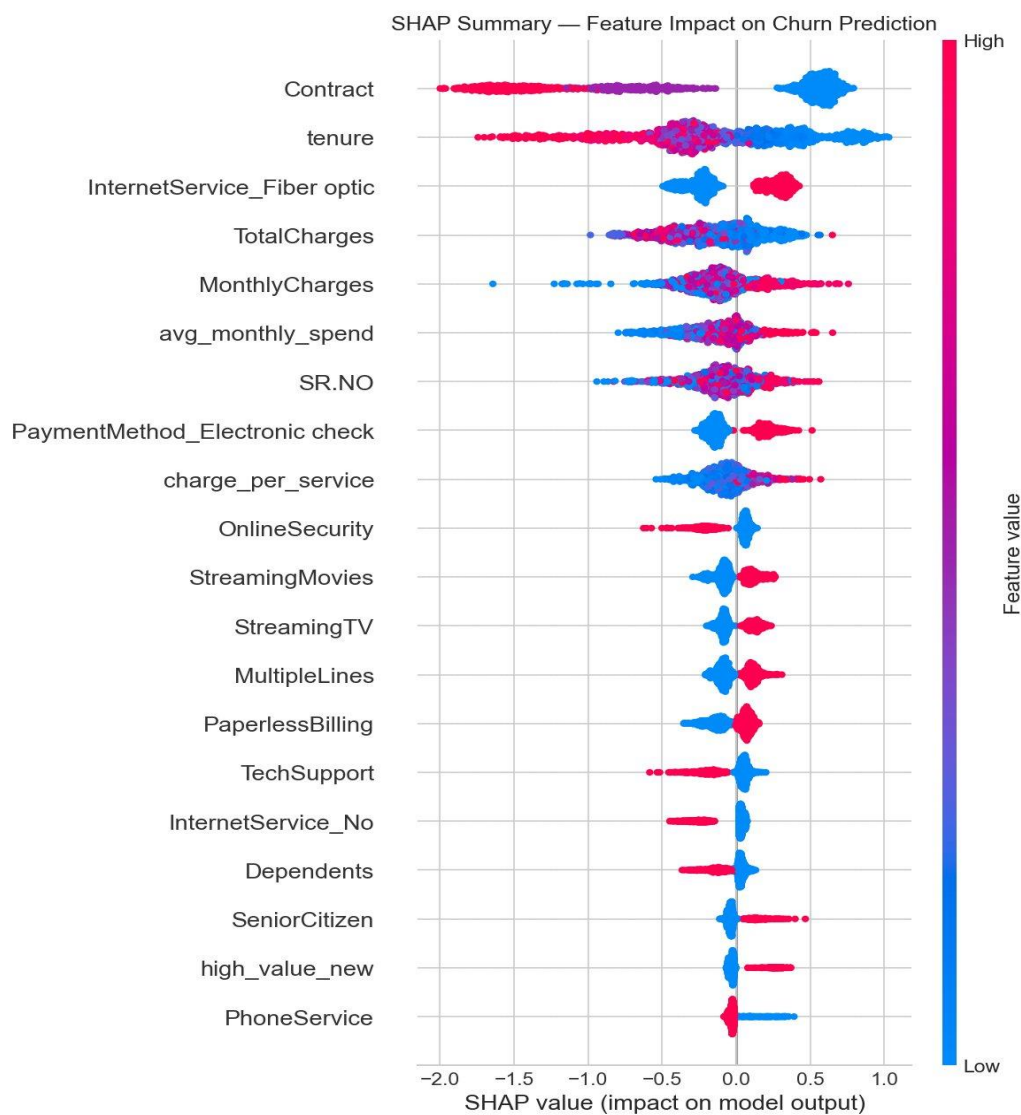


Figure 8: SHAP Summary — feature impact on churn prediction (top 20 features)

4.2 Key Feature Insights from SHAP

Contract Type (Most Important Feature)

Contract type has the largest SHAP values of any feature. Long-tenure blue dots (low encoded contract value = longer contracts) push strongly toward no-churn (negative SHAP), while red dots (month-to-month) push strongly toward churn (positive SHAP). This confirms contract type as the primary lever for retention.

Tenure (Second Most Important)

Long tenure (blue dots) strongly decreases churn probability. Short-tenured customers (red) show positive SHAP values pushing toward churn. The wide spread indicates tenure is highly variable in its effect across customers, making it critical for early intervention targeting.

Fiber Optic Internet Service

Customers with fiber optic internet (red = high value) show elevated churn probability. This may indicate service quality dissatisfaction, higher pricing relative to perceived value, or the fact that fiber customers have more competitive alternatives. This is an operational signal requiring investigation.

Electronic Check Payment

Electronic check users show higher churn rates. Customers on automatic payment methods (credit card or bank transfer) are more likely to stay. This could reflect customer engagement level — customers less invested in the relationship tend to use manual payment methods.

Online Security & Tech Support

Customers without online security or tech support show higher churn. These add-on services may increase perceived value and switching costs. Offering these services as part of retention packages could be an effective strategy.

Monthly Charges & Total Charges

Higher monthly charges (red) moderately increase churn probability. Total charges show an inverse pattern — high total charges (long tenure customers) reduce churn probability. This reinforces the tenure effect: once customers have invested significantly, they are less likely to leave.

5. Business Recommendations

5.1 High-Priority Retention Strategies

Based on the model findings and feature importance analysis, the following retention strategies are recommended in priority order:

Priority	Strategy	Expected Impact
1 — Critical	Incentivize month-to-month customers to upgrade to annual contracts with discounts (10–20%). Target customers flagged as high-risk by the model.	<i>Potential 35–40% churn reduction in this segment</i>
2 — High	Launch early-engagement program for customers in months 1–5. Personalized onboarding, check-in calls, and service reviews.	<i>Address the high early-churn concentration</i>
3 — High	Bundle online security and tech support for fiber optic customers at a subsidized rate. These add-ons significantly reduce churn.	<i>Reduce fiber customer churn from 42% closer to 20%</i>
4 — Medium	Encourage migration from electronic check to auto-pay (credit card or bank transfer). Offer a small monthly discount.	<i>Reduce payment-friction-related churn</i>
5 — Medium	Implement a senior citizen loyalty program with personalized support given their 42% churn rate.	<i>Capture underserved high-risk demographic</i>

Table 4: Prioritized retention strategies based on model insights

5.2 Model Deployment Recommendations

- Deploy the XGBoost model in production with monthly scoring cycles to flag at-risk customers
- Use a churn probability score (0–100) rather than a binary flag to enable tiered intervention intensity
- Integrate with CRM system to trigger automated retention workflows for customers scoring above 60%
- Monitor model performance quarterly and retrain when AUC drops below 0.80
- A/B test retention interventions to measure actual lift and ROI

6. Technical Appendix

6.1 Model Configuration

The following hyperparameters and configurations were used for the final models:

Parameter	Logistic Regression	Random Forest	XGBoost
Class Imbalance	class_weight='balanced'	class_weight='balanced'	scale_pos_weight=2.77
Max Iterations / Estimators	1000	100 trees	100 rounds
Solver / Criterion	lbfgs	gini	logloss
Regularization	C=1.0	max_depth=None	max_depth=6, lr=0.1

Table 5: Model hyperparameter configurations

6.2 Feature Engineering

- avg_monthly_spend: TotalCharges / (tenure + 1) — captures spending rate over time
- charge_per_service: MonthlyCharges / (number of active services) — value-for-money proxy
- high_value_new: Binary flag for customers with high charges and tenure < 12 months — high-risk segment

6.3 Evaluation Metrics Explained

For churn prediction, recall is prioritized over precision because the cost of a missed churner (false negative) exceeds the cost of a false alarm (false positive). The AUC metric provides threshold-independent performance evaluation:

- AUC-ROC: Area Under the Receiver Operating Characteristic Curve — measures discriminative ability across all thresholds. Values > 0.80 are considered good.
- Precision: Of all predicted churners, what fraction actually churned. Higher precision = fewer wasted retention offers.
- Recall: Of all actual churners, what fraction was correctly identified. Higher recall = fewer missed churn events.
- Threshold tuning: Moving from 0.5 to 0.62 reduces false positives at the cost of slightly lower recall — calibrated for this business context.

6.4 Limitations & Future Work

- The dataset does not include customer service interaction logs, which could significantly improve recall
- No temporal data beyond tenure — adding time-series billing patterns could improve model performance
- External factors (competitor pricing, market changes) are not captured

- Future work: implement survival analysis (time-to-churn) for more nuanced retention timing
- Consider ensemble stacking of all three models to marginally improve AUC.

7. Conclusion

This analysis successfully demonstrates that customer churn in the telecom industry is highly predictable using machine learning. The XGBoost model achieves an AUC of 0.843, enabling the business to correctly identify approximately 70% of churners in advance with a well-calibrated decision threshold.

The analysis reveals that churn is not random — it is concentrated in specific, identifiable customer segments: month-to-month contract holders, early-tenure customers, fiber optic internet users without add-on services, and electronic check payers. These segments provide clear, actionable targets for retention programs.

Most importantly: the difference between a 42.7% churn rate (month-to-month customers) and a 2.8% churn rate (two-year contract customers) demonstrates that contract structure is the most powerful retention tool available — and one that is entirely within the company's control.

By deploying this predictive model in production and executing the recommended retention strategies, the company can expect to reduce overall churn rate by an estimated 4–8 percentage points, representing significant revenue preservation and improved customer lifetime value